

SRI AMAN SIMANGGAN, Malaysia

WMO: 960011

Lat: 1.217N

Lon: 111.450E

Elev: 11

StdP: 101.19

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	22.2	20.8	15.5	23.7	21.1	15.8	23.8	4.5	28.1	4.1	28.1	0.7	110	0.280

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.9	35.0	27.3	34.2	27.3	34.0	27.3	28.6	33.1	28.1	32.7	27.8	32.3	1.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	31.7	27.0	22.7	31.5	26.2	21.7	30.6	91.6	31.4	90.0	32.9	88.6	32.7	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.9	3.3	2.8	20.6	36.1	0.7	0.6	20.1	36.6	19.6	37.0	19.2	37.3	18.7	37.8		
			19.7	29.8	0.7	0.7	19.2	30.3	18.8	30.8	18.4	31.2	17.9	31.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	27.3	27.5	28.0	28.3	28.4	28.3	28.1	28.1	27.9	27.8	27.8	27.6
	DBStd	1.02	0.96	0.99	0.94	0.90	0.98	1.04	1.09	1.09	1.07	0.89	0.78	0.85
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6541	535	491	557	548	571	548	562	561	537	552	533	546
	CDD18.3	3499	277	258	298	298	313	298	304	303	287	294	283	288
	CDH23.3	31377	2221	2172	2673	2746	3007	2858	2889	2888	2623	2536	2393	2372
	CDH26.7	12636	768	786	1058	1109	1249	1223	1290	1274	1109	1004	895	870
Wind	WSAvg	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.9	0.9	0.9	1.0
Precipitation	PrecAvg	3359	374	257	277	275	223	199	186	250	282	300	359	384
	PrecMax	4303	525	617	536	407	360	338	374	739	493	459	542	588
	PrecMin	2455	230	87	78	122	88	132	54	70	77	121	191	156
	PrecStd	457	84	129	124	73	85	50	84	154	103	111	87	93
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	34.1	34.2	35.0	35.1	35.2	35.8	35.2	35.1	34.2	34.0	33.9
		MCWB	27.3	27.2	27.3	27.3	27.6	27.4	27.4	27.0	27.2	27.1	27.3	27.8
	2%	DB	32.8	33.0	33.2	34.0	34.1	34.2	34.2	34.2	34.1	33.2	33.1	32.9
		MCWB	27.0	26.8	26.9	27.3	27.6	27.5	27.2	26.9	27.1	26.9	27.2	27.3
	5%	DB	31.9	32.1	32.9	33.1	33.2	33.8	33.9	33.8	33.1	32.9	32.1	32.0
		MCWB	26.7	26.5	26.7	27.0	27.4	27.4	27.1	26.8	26.9	26.8	26.8	26.9
	10%	DB	30.9	31.1	31.9	32.1	32.8	32.8	32.9	32.8	32.1	31.9	31.2	31.1
		MCWB	26.6	26.4	26.5	26.8	27.2	27.1	26.8	26.6	26.7	26.6	26.8	26.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.2	28.4	28.4	28.8	28.7	29.1	28.2	28.6	28.3	28.3	28.5
		MCDB	31.8	32.4	32.8	33.1	33.7	33.4	33.6	33.2	33.3	32.7	32.3	32.5
	2%	WB	27.5	27.4	27.7	27.8	28.2	28.2	28.0	27.7	27.8	27.6	27.6	27.7
		MCDB	31.2	31.5	32.1	32.4	32.9	32.8	33.0	32.6	32.5	32.0	31.7	31.6
	5%	WB	26.9	26.8	27.1	27.4	27.7	27.7	27.3	27.2	27.3	27.1	27.1	27.2
		MCDB	30.7	30.7	31.4	31.8	32.2	32.2	32.2	32.1	31.8	31.4	31.1	31.0
	10%	WB	26.5	26.5	26.6	26.9	27.2	27.2	26.9	26.7	26.8	26.7	26.7	26.7
		MCDB	29.9	30.1	30.6	31.1	31.4	31.5	31.6	31.4	31.1	30.7	30.4	30.3
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.3	8.7	8.8	8.9	9.2	9.6	9.5	9.1	8.7	8.4	8.2
		MCDBR	9.3	9.4	9.8	9.9	10.1	10.7	11.2	11.2	10.5	10.1	9.3	9.2
		MCWBR	4.3	4.1	4.2	4.4	4.6	4.9	5.0	4.9	4.9	4.6	4.4	4.4
	5% WB	MCDBR	8.5	8.6	9.2	9.3	9.4	9.9	10.2	10.1	9.8	9.2	8.8	8.7
		MCWBR	4.2	4.0	4.2	4.4	4.6	4.8	4.9	4.8	4.9	4.5	4.4	4.4
Clear-Sky Solar Irradiance	taub	0.406	0.423	0.427	0.409	0.411	0.409	0.431	0.529	0.578	0.560	0.430	0.402	
	taud	2.482	2.410	2.396	2.449	2.450	2.448	2.380	2.092	1.955	1.998	2.402	2.497	
	Ebn at Noon	922	909	899	895	872	860	846	783	761	781	892	920	
	Edh at Noon	114	124	126	116	112	110	119	164	192	185	122	111	
All-Sky Solar Radiation	RadAvg	4.44	4.68	4.97	5.03	4.93	4.87	4.97	4.97	4.62	4.58	4.68	4.58	
	RadStd	0.32	0.24	0.22	0.23	0.23	0.26	0.25	0.29	0.46	0.56	0.22	0.20	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.25	+0.15	N/A	+0.37	+0.45	+0.35	N/A	N/A	+130	+130

Nomenclature: See separate page